

Full-Stack AI Mammogram Classifier - Technical Report

FAHM Biotechnology Technical Assessment

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Research / demonstration only. Not a medical device. Not approved or intended for clinical use.

1. Summary

End-to-end binary mammogram classifier (Malignant vs Benign), built and deployed inside the one-day budget the brief allows for. The model is an ImageNet-pretrained EfficientNet-B0, fine-tuned on the CBIS-DDSM PNG mirror, and served by a FastAPI inference service behind a Next.js upload UI. Both services are live on free-tier public infrastructure.

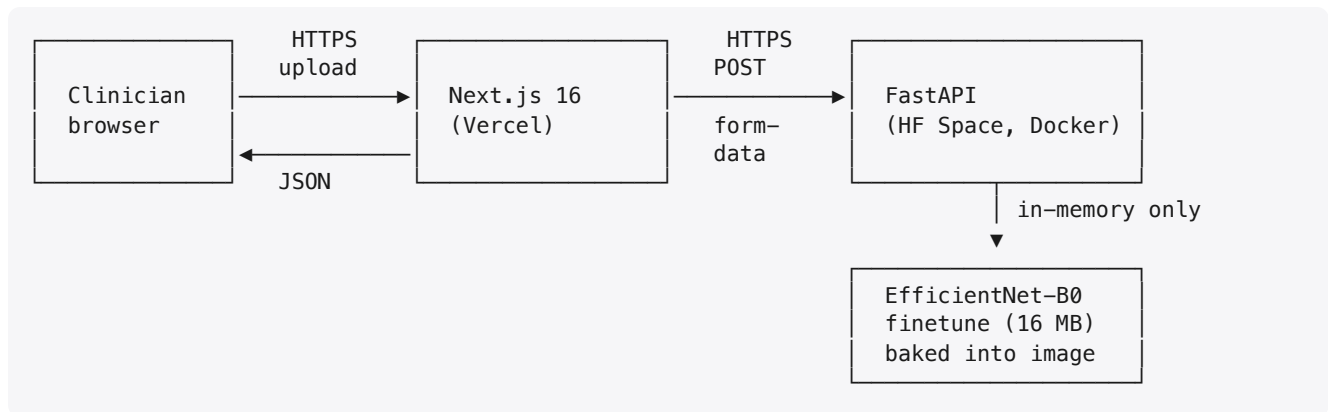
Surface	URL
Frontend (Vercel)	https://mammo-classifier.vercel.app
Backend API (Hugging Face Space, Docker)	https://abwhb-mammo-classifier-api.hf.space
API health	https://abwhb-mammo-classifier-api.hf.space/healthz
Source repository	https://github.com/abwhb/mammo-classifier

Headline metrics on the held-out CBIS-DDSM test set (n = 641, 41 % malignant):

Operating point	Threshold	Sensitivity	Specificity	Accuracy	ROC AUC	PR AUC
Screening (API default)	0.278	0.923	0.391	0.607	0.754	0.671
Balanced (Youden's J)	0.530	0.673	0.659	0.665	0.754	0.671
Diagnostic (≥ 0.80 val spec)	0.614	0.581	0.753	0.683	0.754	0.671
Default sigmoid threshold	0.500	0.708	0.635	0.665	0.754	0.671

The API runs at the **screening** operating point. For a screening tool a missed cancer is far more costly than a callback (see Section 4.2). Bottom line: **sensitivity 0.923 at AUC 0.754**.

2. Architecture



Frontend (apps/web). Next.js 16 App Router with Tailwind v4 and shadcn/ui. A single page composed of a `react-dropzone` (accepts `.dcm`, `.png`, `.jpg`, up to 20 MB), a preview pane, and a result card showing label, confidence, malignancy probability, operating threshold, model version, latency, and request ID. A server-side route handler at `/api/predict` proxies multipart uploads to the API. This hides the backend URL from the client bundle and is the single place to add per-request tracing or a shared-secret header.

Backend (apps/api). Python 3.11 with FastAPI and uvicorn. The full hot path is `privacy.strip_identifiers()` → `ood.is_plausible_mammogram()` → `preprocess.to_tensor()` → `inference.predict()`. The OOD guard (Section 4.4) is a cheap heuristic that catches obvious non-mammogram inputs (screenshots, photos, documents) before they reach the model, which itself has no out-of-distribution awareness. The classifier is loaded once at startup into a module-level `torch.nn.Module` in `eval()` and `no_grad` mode. CPU-only inference latency is roughly 100 to 250 ms per image. The Docker image is 2.0 GB compressed, built with `uv` in a two-stage build (`apps/api/Dockerfile`).

Cloud infrastructure. Backend on a Hugging Face Docker Space (CPU basic, free tier); frontend on Vercel (Hobby tier). The model artifact ships *inside* the Docker image to eliminate the cold-start network dependency. Deployment is a single `infra/deploy_hf_space.py` call (uses `huggingface_hub` directly) plus `vercel deploy --prod`.

3. Dataset and training

3.1 Dataset choice

I chose **CBIS-DDSM** (Curated Breast Imaging Subset of DDSM): a peer-reviewed curation of the canonical DDSM mammography dataset, with binary pathology labels (`MALIGNANT` / `BENIGN`) that map cleanly to the brief's task.

Why CBIS-DDSM rather than MIAS or INbreast?

- **MIAS** has only 322 images digitised from scanned film in the 1990s, with coarse 200 μm pixel resolution and dated labelling conventions. Too small for a meaningful train/val/test split, and the imaging modality (scanned film) is not what a modern screening device produces.
- **INbreast** is higher quality (full-field digital mammography, 70 μm resolution) but contains only 410 images across 115 patients. The licence is also restricted to direct request from the authors, which fits poorly with a publicly-reproducible one-day build.
- **CBIS-DDSM** sits in the middle: 3,103 curated images across 1,566 patients, FFDM-equivalent quality after rescaling, a permissive licence, and a peer-reviewed train/test partition designed to prevent patient leakage. It is the standard public benchmark cited in the medical-imaging literature for this exact task.

For the actual download I used the **PNG mirror at dbaek111/CBIS-DDSM_1024** on Hugging Face Hub: 3,086 1024 $\times$ 1024 8-bit PNGs (638 MB total) derived from the original DICOMs, with the upstream patient-disciplined train/test partition preserved as a folder layout.

Why this mirror over the canonical TCIA source? The TCIA download is roughly 163 GB of DICOMs and does not fit the one-day budget. The PNG mirror trains in 10 to 20 minutes on a single Apple Silicon laptop, and the upload UI still accepts raw DICOMs in production via the API's `pydicom`-based preprocessing path (Section 3.4). The trade-off is recorded in Section 6.

3.2 Splits

CBIS-DDSM's authors publish a patient-disciplined train/test partition (the same patient never appears in both train and test). The PNG mirror preserves that partition as `train/` and `test/` folders. I keep their test set untouched and carve an 18 % **stratified-random** validation set out of `train/`. The mirror named files by SOP Instance UID rather than patient ID, so the train/val carve cannot be patient-grouped from filenames alone. The val set may therefore contain images from patients also represented in train, which would slightly inflate the val metric. This does **not** affect the held-out test metric, which is what is reported below.

Split	Images	Benign	Malignant	Prevalence (malignant)
Train	2,004	1,104	900	0.449
Val	441	243	198	0.449
Test	641	381	260	0.406

3.3 Model and training recipe

- **Architecture.** `torchvision.models.efficientnet_b0` with ImageNet-1k pretrain. The final classifier `Linear(in_features, 1000)` is swapped for `Linear(in_features, 1)`; the output is a logit, with sigmoid applied at the operating-point boundary.
- **Loss.** `BCEWithLogitsLoss(pos_weight=...)` with `pos_weight = neg/pos  $\approx$  1.23` to counter the mild class imbalance.
- **Optimizer.** AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-4} , cosine schedule over 10 epochs.

- **Augmentation.** `Resize` → `RandomResizedCrop(224, scale=0.85–1.0)` → `RandomHorizontalFlip(p=0.5)` → `RandomRotation( $\pm 10^\circ$ )` → `ImageNet Normalize`. **No vertical flips:** mammogram views (CC/MLO) have consistent vertical orientation that flipping would corrupt.
- **Early stopping.** Track val AUC; stop after three epochs without improvement.
- **Device.** Auto-detect CUDA → MPS → CPU. Trained locally on Apple Silicon MPS in about 9 minutes.

Result. Best val AUC **0.789** at epoch 9 (full history in `reports/train_history.json`).

3.4 Inference preprocessing

The API accepts DICOM, PNG, and JPEG. For DICOM it parses with `pydicom`, applies the VOI LUT if present, inverts MONOCHROME1 series, and converts to float32. For PNG/JPEG it reads via Pillow as L (grayscale). All formats then go through a common pipeline: 1st/99th percentile clip, min-max normalisation to $[0, 1]$, resize to 224×224 (bilinear), tile the single channel to 3 (EfficientNet expects 3-channel input), and ImageNet mean/std normalisation.

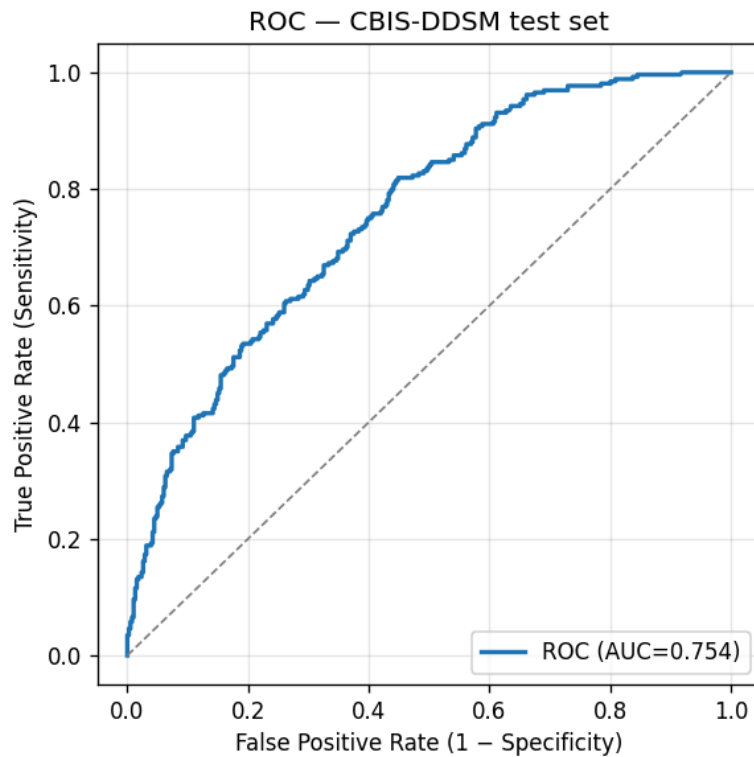
4. Evaluation

4.1 Why accuracy alone is insufficient

The CBIS-DDSM test set is 41 % malignant. In a real screening cohort the prevalence of malignancy is closer to 0.5 to 1 %. A constant "benign" predictor would score 99 % accuracy in that setting while catching zero cancers. **Accuracy is unreadable without prevalence.** For a screening task the metrics that matter are:

- **Sensitivity (recall on malignant):** fraction of actual cancers the model catches.
- **Specificity:** fraction of benign cases the model leaves uncalled.
- **ROC AUC:** threshold-free measure of discriminative power, independent of operating point and prevalence.
- **PR AUC:** more informative than ROC AUC at low malignant prevalence, which is the deployed setting one would actually care about, not the curated 41 % test set.

The model achieves **ROC AUC 0.754** on the held-out test set. PR AUC is **0.671**. Both reflect a useful but not state-of-the-art classifier; published deep models on full-DICOM CBIS-DDSM with in-domain pretraining reach 0.85 to 0.95 AUC. The gap is mostly explained by the trade-offs table in Section 6.



4.2 Threshold selection and the three operating points

The default 0.5 sigmoid threshold is rarely the right operating point in medical imaging. Lowering it raises sensitivity at the cost of specificity, and the right place to sit on that curve depends on the use case. I report three thresholds chosen on the **validation** set and evaluated on the **test** set, plus the default 0.5 for transparency:

Operating point	Threshold	Val rationale	Test sens	Test spec	Test acc
Screening (API default)	0.278	smallest threshold with val sensitivity ≥ 0.90	0.923	0.391	0.607
Balanced	0.530	maximises Youden's J on val	0.673	0.659	0.665
Diagnostic	0.614	smallest threshold with val specificity ≥ 0.80	0.581	0.753	0.683
Default 0.5	0.500	sigmoid default, reported for reference	0.708	0.635	0.665

The **API uses the screening operating point**. Concretely, the test-set confusion matrix at that threshold is:

	Predicted benign	Predicted malignant	Row total
Actual benign (381)	TN = 149	FP = 232	381
Actual malignant (260)	FN = 20	TP = 240	260
Column total	169	472	641

In plain terms: out of 260 malignant cases in the test set, the model catches 240 and misses 20. Out of 381 benign cases it calls back 232 for follow-up. For a screening tool that is the right trade-off, since a false negative may be a missed diagnosis and a false positive triggers an additional read, which is recoverable. A diagnostic tool would sit at a different point on the same curve.

4.3 Class imbalance

The CBIS-DDSM test set is 41 % malignant, far higher than a real screening cohort (around 0.5 to 1 %). The model was trained with `BCEWithLogitsLoss(pos_weight=1.23)` to account for the mild train-set imbalance (55 % benign), but training-time imbalance is not the larger story. The larger story is that **the metrics shown above will degrade in production** if deployed on a real screening population, because the same operating point will produce a much higher false-positive count relative to the now-rare true positives. PR AUC, which I report alongside ROC AUC, is more sensitive to this prevalence shift and is the metric to watch in deployment monitoring.

4.4 Out-of-distribution inputs

The classifier was trained only on mammograms and has no calibrated way to say "this isn't a mammogram." Fed an arbitrary photo, screenshot, or document, the sigmoid head still produces a probability, and at the aggressive screening threshold of 0.278 most non-mammogram inputs will land above it. This produced a sharp early failure mode in informal testing: a dark-themed chat screenshot was confidently flagged "malignant".

The mitigation in `apps/api/app/ood.py` is a cheap two-feature heuristic that runs after DICOM de-identification and before preprocessing:

1. **Dark-background fraction.** Mammograms in standard CC/MLO views sit on a near-black background. At least 15 % of pixels below intensity 0.12 is required.
2. **Edge density.** Soft tissue is smooth; UI, text, and diagrams are not. Inputs with more than 22 % of pixels flagged by PIL's `FIND_EDGES` filter are rejected.

Calibration was done by hand against a handful of CBIS-DDSM test images and three OOD synthetics (uniform photo, dark-theme screenshot, uniform random noise). All five CBIS-DDSM samples I checked pass (dark fraction 0.43 to 0.66, edge density 0.01 to 0.04, well inside the bounds); all three OOD synthetics are rejected with a 422. DICOM inputs bypass the check since the format itself is strong evidence of intent.

This is not a real OOD detector. A production system needs either a dedicated mammogram-vs-other classifier or density-based novelty detection in feature space. The heuristic catches the obvious cases but would not reliably reject a high-resolution photograph of a chest X-ray, for instance. It is listed in the trade-offs table in Section 6.

5. Data privacy and security framework

Mammograms are sensitive health data. The submission is a research demo, not a regulated medical device, but the architecture is built to give the privacy posture real substance, not just disclaimers.

5.1 In transit

- All public surfaces are HTTPS only.
- Vercel terminates TLS 1.3 at the edge for the frontend, with HSTS on.
- Hugging Face Spaces gives every Docker Space a managed-TLS `*.hf.space` domain.
- The browser never talks to the backend directly: the Vercel route handler `/api/predict` proxies the upload server-to-server, so the backend URL is not exposed in the client bundle and is the only place to add per-request tracing or a shared-secret header.

5.2 At rest

- **Uploaded images are never written to disk in production.** FastAPI reads the upload into memory, processes it, and discards it after the response. There is no tempfile, no upload bucket, no caching layer.
- **The model artifact** is baked into the Docker image and lives in immutable container storage; it contains no PHI.
- **Logs** are structured JSON containing only: request ID (UUID v4, generated server-side), file size, MIME type, inference latency, predicted label, malignancy probability, model version, and operating threshold. No filename, no DICOM headers, no image bytes, no PHI.

5.3 De-identification for DICOM specifically

`apps/api/app/privacy.py:strip_identifiers` runs **before** preprocessing on any DICOM upload (PNG and JPEG carry no DICOM PHI tags). It implements a subset of the DICOM Basic Application Confidentiality Profile (DICOM PS 3.15 Annex E, Basic Profile):

1. Removes every direct-PHI tag in `_PHI_TAGS_TO_REMOVE`: patient name, ID, birth date, sex, age, address, telephone, ethnic group, accession number, every physician name, institution name, address, department, station name, all date/time fields, study and series descriptions, other patient IDs, device serial number, and similar.
2. Rotates the Study, Series, and SOP Instance UUIDs to fresh values. These can be PHI when the originals encode site-specific patterns; they must remain present for the file to stay DICOM-valid.
3. Strips every private tag block via `ds.remove_private_tags()`. Vendor-specific tags often carry PHI not covered by the standard tag list.
4. Recurses into nested sequences (for example `ReferencedStudySequence`) and scrubs them too.

This is unit-tested by constructing a synthetic DICOM populated with every tag in the removal list (including a private vendor tag block), running the scrubber, and asserting that every PHI tag is gone, every UID is rotated, and no private tags survived (`apps/api/tests/test_privacy.py`).

5.4 Saudi PDPL alignment

The Personal Data Protection Law (Royal Decree M/19, in force 14 September 2023) governs personal-data processing in Saudi Arabia. The relevant principles and how this design addresses them:

PDPL principle	How this design addresses it
Lawful basis, purpose limitation (Arts. 5 to 6)	The app processes uploads only to produce a single inference response. No secondary use, no analytics on the image, no third-party telemetry on the upload pipeline.
Data minimisation (Art. 11)	The pipeline extracts pixels and discards everything else. No user accounts, no metadata beyond non-identifying request telemetry.
Storage limitation (Art. 18)	Images are never persisted. Logs retain only non-identifying metadata.
Security of processing (Art. 19)	TLS everywhere; backend not directly browser-callable; container runs as non-root uid 1000; principle-of-least-privilege image.
Cross-border transfer (Art. 29)	For this demo the backend runs on Hugging Face Spaces, which terminates in EU and US datacenters. This is an explicit gap for KSA data residency. The intended production target is GCP Cloud Run in <code>me-central1</code> (Dammam) for in-Kingdom processing; the Dockerfile is portable and would deploy there unchanged.
Data subject rights (Arts. 21 to 26)	Since the system stores neither identifiers nor images, the access, rectification, and erasure rights are satisfied by design: there is nothing to access or erase.
Breach notification (Art. 20)	Not applicable in the demo (no real PHI is processed). Listed in Section 6 as a "with more time" item.

Two gaps to flag plainly: **(1)** Hugging Face Spaces is not in-Kingdom; for real KSA-resident PHI the backend needs to move to a KSA-region host (GCP `me-central1` , STC Cloud, or self-hosted). **(2)** This is an *alignment* claim, not a compliance certification. A real deployment would need legal review of the same controls.

5.5 In-product disclaimers

Every screen of the UI carries the line "Research / demonstration only - not for clinical use." The API response includes the same string in the `disclaimer` field.

6. Trade-offs and what I would do with more time

This section is the one the brief explicitly asks for. The following are the trade-offs I made under the one-day budget, and what I would change first if given a week.

Decision (this build)	Trade-off	If I had more time
HF PNG mirror, not full TCIA DICOMs	About 5× less data, lower fidelity than full-bit-depth source	Full CBIS-DDSM DICOMs with 16-bit pixel data and proper VOI LUT handling
EfficientNet-B0 (ImageNet pretrain)	Domain gap: natural images → mammograms	Pretrain on a large in-domain corpus (RSNA Mammography, EMBED) before CBIS-DDSM fine-tune
224 × 224 input	Loses fine calcification detail visible at full resolution	Multi-scale patch ensemble at 512 × 512 or 1024 × 1024, with attention over patches
Image-level labels only	Ignores standard 4-view fusion (L/R × CC/MLO)	Per-patient model that fuses all four standard views with view-aware pooling
Stratified random val split	Cannot guarantee no patient overlap with train (no patient IDs in HF mirror filenames)	Re-join with TCIA metadata CSVs to recover patient IDs, then <code>GroupShuffleSplit</code>
Single operating threshold	Hides the precision/recall trade space from the user	Interactive ROC in the UI; user picks the operating point per use case
No CI/CD	Manual <code>vercel deploy plus python infra/deploy_hf_space.py</code>	GitHub Actions: lint and tests on PR; deploy on main; preview URLs
No deployment monitoring	Drift is invisible until a user complains	Log probability distribution, alert on shift; weekly held-out re-evaluation
No authentication	API is publicly callable	OAuth (Clerk or Auth0) plus per-clinician audit log, signed upload URLs
HF Space EU/US region	Not in-Kingdom for KSA PHI	Production target is GCP Cloud Run in <code>me-central1</code>
Heuristic OOD guard	Catches obvious non-mammograms but not subtle ones	Dedicated mammogram-vs-other classifier or feature-space novelty detection
In-memory inference only	No traceability for audit	Encrypted, time-bounded audit storage with patient-consent gating; PDPL Art. 20 breach-notification runbook

7. Acknowledgements and licensing

- **Dataset.** Lee R. S., Gimenez F., Hoogi A., Miyake K. K., Gorovoy M., Rubin D. L. *A curated mammography data set for use in computer-aided detection and diagnosis research*. Scientific Data 4:170177 (2017). Released by TCIA under **Creative Commons Attribution 3.0 (CC BY 3.0)**. PNG mirror by `dbaek111` on Hugging Face Hub inherits the same licence.
- **Architecture.** Tan M., Le Q. V. *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. ICML 2019.
- **Tooling.** PyTorch, FastAPI, pydicom, Next.js 16, shadcn/ui, `uv`, Hugging Face Hub, Vercel.